

Engineering Notes

Arcsine Vector Field for Path Following Guidance

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I. Introduction

UNMANNED aerial vehicles (UAVs) find numerous applications in real-world scenarios, such as agriculture, healthcare, disaster management, defense, security, and delivery. In that context, UAVs are often desired to autonomously follow predefined paths. Path following algorithms should cater to the dynamic capabilities of UAVs and provide accurate performance while utilizing, preferably, a computationally inexpensive guidance logic.

Vector-field-based path following algorithms provide an elegant framework where path planning and tracking elements are fused into one. The approach uses a position-based guidance command to provide a unique heading direction at every point in the workspace that guides the UAV onto the desired path. Accordingly, the guidance input in the vector field method is resilient to position disturbances and effect of vehicle dynamics. Vector-field-based path following algorithms also offer a promising solution for achieving low path following errors [1]. The existing literature on the vector field approach can be broadly characterized into two, namely, gradient-based methods [2–10] and direct methods [11–17]. In gradient-based methods, the commanded heading direction is deduced from the gradient of a potential function that is constructed to have its minimum on the desired path, whereas direct methods use the current position of the UAV relative to the path to directly determine the desired heading direction. Frew et al. [2] introduced a Lyapunov vector field method by deducing the gradient of the potential function based on a circular path. Therein, the guidance command comprises a contraction term that governs the radial speed, and a circulation term that controls the tangential speed with respect to the desired path. A hybrid vector field strategy to follow a circular path is presented in Ref. [3], wherein, for the initial position outside the circle, UAV follows the shortest path along the tangent to the desired circle, and the vector field method used in Ref. [2] is used while traversing from inside the circle. Lim et al. [4] modified the vector field method used in Ref. [2] by introducing a guidance parameter that is iteratively calculated to vary the speed of UAVs for achieving simultaneous arrival on a desired circle. By constructing the circulation parameter as a function of radial distance, Pothan and Ratnoo [5] presented a vector field method that settles faster onto the desired circular path while requiring a lower maximum curvature. The Lyapunov vector field method is further extended to track a dynamic star curve in

Ref. [6]. Sun et al. [7] modified the circulation parameter introduced in Ref. [5] for improving the settling time characteristics. The authors of Refs. [8,9] use the basic idea of computing the desired heading direction from the vector sum of tangential and normal components defined with respect to a desired path. Those two methods differ in construction of their tangential components contributing to the vector field. Goncalves et al. [8] derive the tangential component as the cross product of the gradients of the two functions, the intersection of which generates the desired path. In contrast, Kapitanyuk et al. [9] directly obtain the tangential component by rotating the normal vector. Wilhelm and Clem [10] combine the path following method in Ref. [8] with a weighted repulsive vector field to avoid circular obstacles. Gradient-based methods require the computation of gradients for deducing the guidance command, which adds to the computational requirement. Unlike these methods, the authors of Refs. [11–14] present a class of vector fields where UAV heading direction can be uniquely and directly determined by the UAV's relative position with respect to the desired path. Accordingly, direct methods are simple, computationally efficient, and easily implementable forms of vector field guidance logic. In Ref. [11], the heading guidance command is derived as an inverse tangent function of the UAV position error with respect to a straight-line or circular path. By using the tangent direction at a reference point on the desired path, Griffiths [12] modified the guidance law used in Ref. [11] to follow a sinusoidal path. Therein, the instantaneous reference point on path has the same abscissa as that of the UAV. Shivam and Ratnoo [13] discuss arcsine-function-based commanded course angle variation for following an elliptic path. The authors of Refs. [15,16] focus on implementing the vector field method in Ref. [11] for scenarios with unmodeled UAV dynamics and unknown wind components. Wang et al. [17] use the guidance logic presented in Ref. [11] for follower UAV course angle and ground speed control in leader–follower formations.

The main contribution of this work is a novel arcsine vector field for UAV path following guidance. The proposed direct vector field method uses an arcsine function to determine the commanded course angle based on the UAV's position error relative to a straight-line or circular path. The study also includes a curvature analysis, resulting in a closed-form expression for the maximum curvature for the straight-line path following case. By comparing the proposed method with a popular vector field [11], it was found that the maximum curvature and control effort were significantly reduced. Furthermore, the research highlights the usefulness of the proposed guidance method by extending it to variable curvature paths. While sample simulation results were presented in a previous work [14], no validation studies were conducted in the presence of wind and autopilot dynamics or variable curvature paths. This work fills those gaps and establishes the merit of the proposed guidance method through rigorous comparative analysis.

The remainder of this Note is organized as follows: The problem statement is defined in Sec. II. Sections III and IV discuss the straight-line and circular path following scenarios, respectively. Section V presents simulation studies, and Sec. VI concludes this paper.

II. Problem Statement

Consider straight line and circular orbit as two representative path following scenarios as shown in Fig. 1. In Fig. 1a, UAV is at an instantaneous lateral or cross-track error x , from the desired straight-line path that, without loss of generality, is considered along the y axis. Here, V_g and χ denote UAV ground speed and course angle, respectively. The objective here is to develop a prospective vector field guidance law that governs the UAV course angle and drives the cross-track error to zero. In the case of following a circular path as

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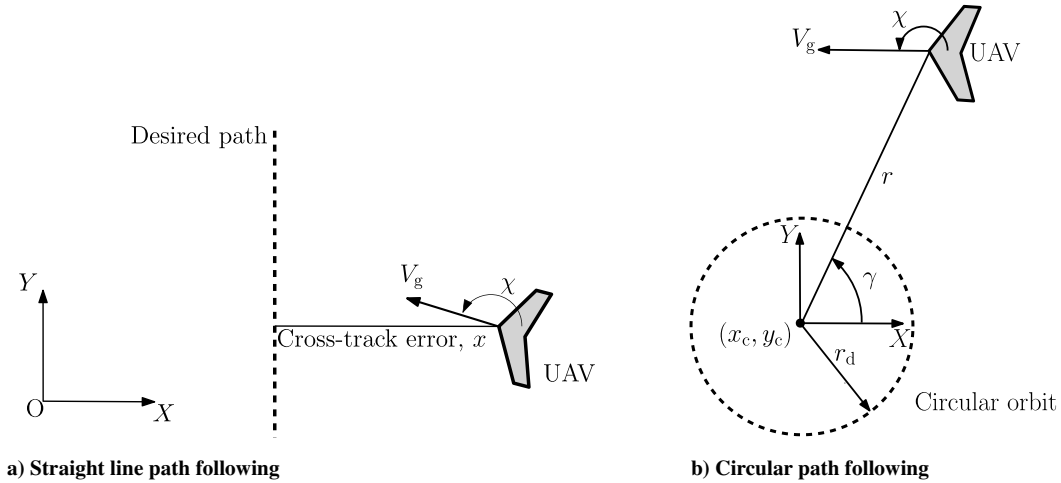


Fig. 1 Path following scenarios. (Figures adapted from Ref. [14] with permission from IEEE.)

shown in Fig. 1b, a similar objective is to govern χ for settling on to the desired circular orbit of radius r_d . The UAV instantaneous position, in this case, can be characterized by the radial distance r from the centre of the circle (x_c, y_c) and the orientation angle γ as defined in a fixed frame.

The kinematic equations of motion of the UAV can be described as

$$\dot{x} = V_g \cos \chi \quad (1)$$

$$\dot{y} = V_g \sin \chi \quad (2)$$

III. Straight-Line Following

This section discusses the proposed vector field guidance logic and its properties for the straight path following problem.

A. Proposed Guidance Law

Commanded course angle χ_d for the UAV is proposed as

$$\chi_d = \begin{cases} \sin^{-1}\left(\frac{1}{1 + k_s x^2}\right), & x \leq 0 \\ \pi - \sin^{-1}\left(\frac{1}{1 + k_s x^2}\right), & x > 0 \end{cases} \quad (3)$$

where k_s is the guidance gain that can take positive values. The proposed arcsine function offers a smooth and bounded variation in the course angle. Further, the function is analytically tractable, which leads to a feasible curvature analysis. Figure 2a illustrates the variation of commanded course angle with respect to the cross-track error

for $k_s = 0.002$. It can be seen that the commanded course angle varies smoothly for $x \in (-\infty, \infty)$ and at $x = 0$ it prescribes $\chi_d = \pi/2$, which is the desired direction of motion along the path. For very large x , that is, $x \gg 0$, $\chi_d \rightarrow \pi$, which is the direction along the shortest distance to the path. Similarly, for $x \ll 0$, $\chi_d \rightarrow 0$. Overall, the proposed arcsine shaping provides smooth variation in course angle guiding the UAV to the path. The commanded course angle variation for different values of k_s is plotted in Fig. 2b. As shown in that figure, higher values of k_s lead to steeper variation in course angle near the desired path. Further, as will be discussed in the following subsection, the guidance gain k_s provides a design flexibility in abiding by the maximum turn-rate constraint of the UAV.

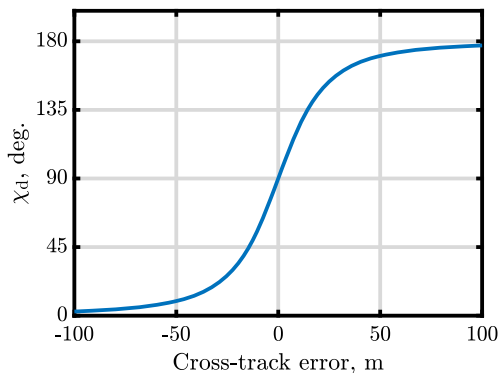
B. Curvature Characteristics

This subsection analyzes the curvature variation for the proposed vector field and deduces a closed-form expression for the maximum curvature. Further, a comparison of curvature characteristics is carried out with the vector field presented in Ref. [11]. Curvature κ is related to the course rate through the speed of UAV V_g as

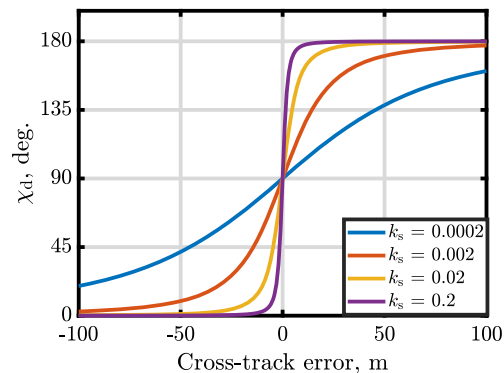
$$\kappa = \frac{\dot{\chi}_d}{V_g} \quad (4)$$

Differentiating Eq. (3) with respect to time leads to

$$\dot{\chi}_d = \begin{cases} -\frac{2k_s}{(1 + k_s x^2)\sqrt{k_s^2 x^2 + 2k_s}} \dot{x}, & x \leq 0 \\ \frac{2k_s}{(1 + k_s x^2)\sqrt{k_s^2 x^2 + 2k_s}} \dot{x}, & x > 0 \end{cases} \quad (5)$$



a) Course angle variation for $k_s = 0.002$



b) Course angle variations for different k_s

Fig. 2 Commanded course angle profiles.

Using Eqs. (1) and (3),

$$\dot{x} = \begin{cases} V_g \cos\left(\sin^{-1}\left(\frac{1}{1+k_s x^2}\right)\right), & x \leq 0 \\ -V_g \cos\left(\sin^{-1}\left(\frac{1}{1+k_s x^2}\right)\right), & x > 0 \end{cases} \quad (6)$$

Using Eqs. (5) and (6), the course rate can be deduced as

$$\dot{\chi}_d = -\frac{2V_g k_s x}{(1+k_s x^2)^2} \quad (7)$$

Using Eqs. (4) and (7), curvature can be expressed as

$$\kappa = -\frac{2k_s x}{(1+k_s x^2)^2} \quad (8)$$

Proposition 1: With the course angle of the UAV governed by Eq. (3), the magnitude of maximum curvature κ_m along the trajectory is given by

$$|\kappa_m| = \begin{cases} \frac{9}{8} \sqrt{\frac{k_s}{3}}, & |x_0| \geq \frac{1}{\sqrt{3k_s}} \\ \frac{2k_s |x_0|}{(1+k_s x_0^2)^2}, & |x_0| < \frac{1}{\sqrt{3k_s}} \end{cases} \quad (9)$$

Proof: Differentiating Eq. (8) with respect to x leads to

$$\frac{d\kappa}{dx} = \frac{-2k_s[1-3k_s x^2]}{(1+k_s x^2)^3} \quad (10)$$

Using Eq. (10), $(d\kappa/dx) = 0$ for

$$x = \pm \frac{1}{\sqrt{3k_s}} \quad (11)$$

Three cases are considered based on the nature of the UAV initial position.

Case 1: $x_0 < 0$

Evaluating Eq. (8) at $x = -(1/\sqrt{3k_s})$ gives

$$\kappa|_{x=-1/\sqrt{3k_s}} = \frac{9}{8} \sqrt{\frac{k_s}{3}} \quad (12)$$

Further using Eq. (10),

$$\frac{d\kappa}{dx} > 0 \quad \forall x < -\frac{1}{\sqrt{3k_s}} \quad \text{and} \quad \frac{d\kappa}{dx} < 0 \quad \forall x > -\frac{1}{\sqrt{3k_s}} \quad (13)$$

Since, using Eq. (6), $\dot{x} > 0 \quad \forall x < 0$, Eqs (8), (12), and (13) imply

$$\kappa_m = \begin{cases} \frac{9}{8} \sqrt{\frac{k_s}{3}}, & x_0 \leq -\frac{1}{\sqrt{3k_s}} \\ -\frac{2k_s x_0}{(1+k_s x_0^2)^2}, & -\frac{1}{\sqrt{3k_s}} < x_0 < 0 \end{cases} \quad (14)$$

Case 2: $x_0 > 0$

Using Eq. (8),

$$\kappa(-x) = -\kappa(x) \quad (15)$$

which implies that $\kappa - x$ curve is odd symmetric in nature. Accordingly, using Eqs. (14) and (15), it can be readily deduced that

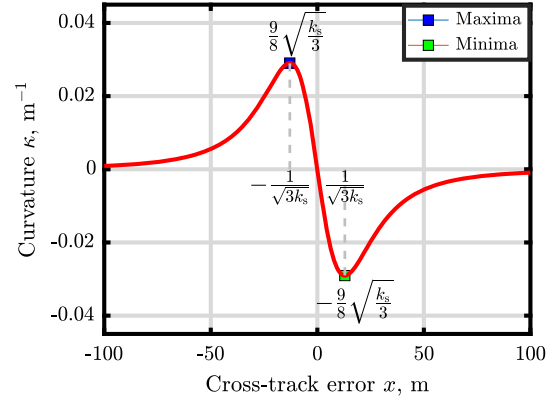


Fig. 3 Curvature variation for $k_s = 0.002$.

$$\kappa_m = \begin{cases} -\frac{2k_s x_0}{(1+k_s x_0^2)^2}, & 0 < x_0 < \frac{1}{\sqrt{3k_s}} \\ -\frac{9}{8} \sqrt{\frac{k_s}{3}}, & x_0 \geq \frac{1}{\sqrt{3k_s}} \end{cases} \quad (16)$$

Case 3: $x_0 = 0$ presents the trivial case where the UAV is already on the desired path following a straight course with $\kappa = 0$. ■

Figure 3 illustrates the curvature variation with cross-track error and highlighting the extreme curvature κ_m and the points of curvature extrema for a guidance gain $k_s = 0.002$. The asymptotic behavior of the guidance law can be verified using direct method of Lyapunov as discussed in Proposition 2.

Proposition 2: Subject to the guidance law given in Eq. (3), the cross-track error x converges asymptotically to zero for any initial condition $x_0 \in (-\infty, \infty)$.

Proof: Consider a positive definite Lyapunov function $W(x) = x^2/2; \mathbb{R} \rightarrow \mathbb{R}$, such that $W(0) = 0$ and $W(x) > 0$ for all $x \in (-\infty, \infty)$. Differentiating W with respect to time leads to

$$\dot{W} = x\dot{x} \quad (17)$$

Using Eqs. (1) and (17),

$$\dot{W} = xV_g \cos \chi_d \quad (18)$$

Using Eqs. (3) and (18),

$$\dot{W} = \begin{cases} xV_g \cos\left(\sin^{-1}\left(\frac{1}{1+k_s x^2}\right)\right), & x \leq 0 \\ xV_g \cos\left(\pi - \sin^{-1}\left(\frac{1}{1+k_s x^2}\right)\right), & x > 0 \end{cases} \quad (19)$$

which leads to

$$\begin{aligned} \dot{W} &= -|x|V_g \cos\left(\sin^{-1}\left(\frac{1}{1+k_s x^2}\right)\right) \\ \Rightarrow \dot{W} &< 0 \quad \text{for all } x \neq 0, \text{ and } \dot{W} = 0 \quad \text{for } x = 0 \end{aligned} \quad (20)$$

■

IV. Circular Path Following

Considering a circular path following scenario, the proposed guidance law and its essential features are discussed. From Fig. 1b, the UAV kinematic equations of motion in polar form can be written as

$$\dot{r} = V_g \cos(\chi - \gamma) \quad (21)$$

$$\dot{\gamma} = \frac{V_g}{r} \sin(\chi - \gamma) \quad (22)$$

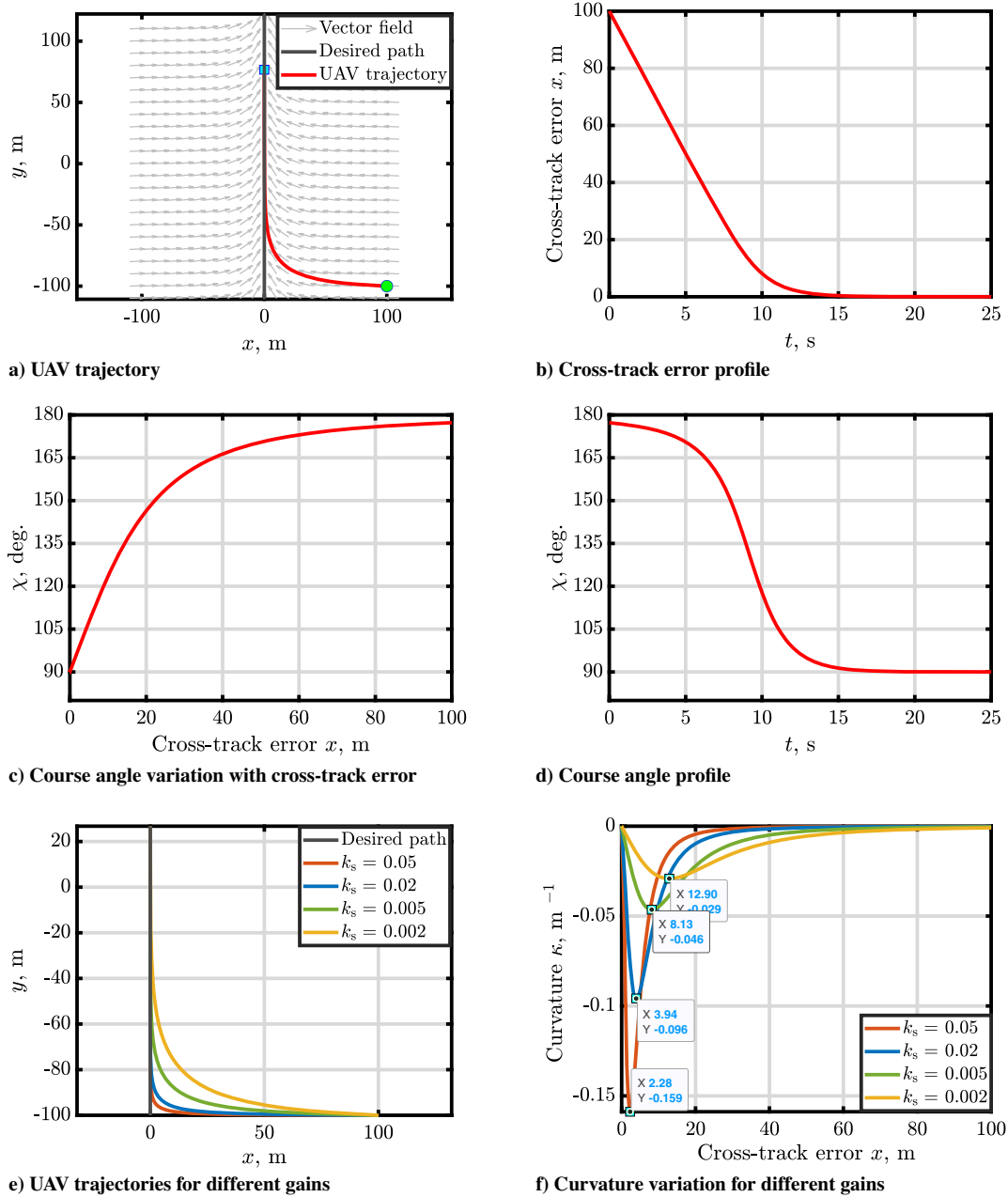


Fig. 5 Results for case 1: straight-line path following.

Table 1 Simulation parameters for straight-line path following using different guidance gains

Initial position, (x_0, y_0) , m	Guidance gain (k_s)	Initial course angle (χ_0) , deg	Point of curvature extrema (x_m) , m	Extreme curvature (κ_m) , m^{-1}
(100, -100)	0.05	179.9	2.28	-0.159
	0.02	179.7	3.94	-0.096
	0.005	178.9	8.13	-0.046
	0.002	177.3	12.90	-0.029

cross-track error reduces to zero. With respect to time, the course angle variation is shown in Fig. 5d. Additional simulations are carried out using different values for the guidance gain k_s as listed in Table 1. Figure 5e shows the UAV trajectories for different guidance gains. The corresponding variation in curvature with cross-track error is shown in Fig. 5f. The extreme curvature and corresponding point of extrema obtained are listed in Table 1, which abide by Eqs. (9)

and (11), respectively. It can be seen from Fig. 5f that, for a given initial condition, larger guidance gain leads to higher extreme curvature as governed by Eq. (9).

The proposed guidance method is compared with the path following technique used in Ref. [11], the commanded course angle for which is given by

$$\chi_d = \frac{\pi}{2} + \tan^{-1}(\hat{k}_s x) \quad (34)$$

where $\hat{k}_s$ is the guidance gain. While studying the relative performance of the two methods, a fair basis of a comparison is needed. The guidance gain for Ref. [11] is computed such that the desired course angle at the initial position matches the one given by the proposed method. Identical initial course angles ensure that the desired direction of motion of the UAV is consistent at the start of the trajectory. Accordingly, considering $x_0 > 0$, the gain $\hat{k}_s$ is computed using Eqs. (3) and (34) as

$$\pi - \sin^{-1}\left(\frac{1}{1 + k_s x_0^2}\right) = \frac{\pi}{2} + \tan^{-1}(\hat{k}_s x_0) \quad (35)$$

which leads to

$$\hat{k}_s = \sqrt{k_s^2 x_0^2 + 2k_s} \quad (36)$$

Differentiating Eq. (34) with respect to time gives course rate $\dot{\chi}_d$, and using Eq. (4), the curvature expression for the vector field method used in Ref. [11] is deduced as

$$\hat{\kappa} = -\frac{\hat{k}_s^2 x}{(1 + \hat{k}_s^2 x^2)^{3/2}} \quad (37)$$

Subsequently, solving $(d\hat{\kappa}/dx) = 0$ leads to the absolute maximum curvature for the vector field given in Ref. [11] as

$$|\hat{\kappa}|_m = \frac{2\hat{k}_s}{3\sqrt{3}} \quad \text{at } x = \pm \frac{1}{\sqrt{2}\hat{k}_s} \quad (38)$$

The initial position of the UAV is again considered as $(x_0, y_0) = (100 \text{ m}, -100 \text{ m})$ with $k_s = 0.005$. Using $k_s = 0.005$ in Eq. (36) leads to the corresponding guidance gain for Ref. [11] as $\hat{k}_s = 0.510$. The initial course angle is along the desired course, which is computed considering $k_s = 0.005$ using Eq. (3) for the proposed method. For Ref. [11], substituting $\hat{k}_s = 0.510$ in Eq. (34) yields $\chi_{d_0} = \chi_0 = 178.88^\circ$. The autopilot gains are chosen as $(k_\chi, k_{\dot{\chi}}) = (200, 14.14)$. The speed of the UAV is considered as $V_g = 10 \text{ m/s}$. Figure 6 illustrates the comparative simulation results. Figure 6a plots the UAV trajectories and the corresponding variation in curvature with cross-track error is shown in Fig. 6b. It can be seen that the maximum curvature for the proposed method is $|\kappa_m| = 0.046 \text{ m}^{-1}$ at $x = 8.16 \text{ m}$, which is significantly lower than $|\hat{\kappa}_m| = 0.216 \text{ m}^{-1}$ at $x = 1.19 \text{ m}$ obtained using Ref. [11]. Figure 6c shows the course

angle variation with cross-track error, and the corresponding course angle profile is plotted in Fig. 6d. Furthermore, to analyze the nature of arcsine-based commanded course angle [Eq. (3)] for the proposed method and the arctan-based commanded course angle [Eq. (34)] of Ref. [11], simulation results are presented for a wide range of guidance gains in the interval $k_s \in [0.1, 10] \times 10^{-3}$ as shown in Fig. 7. Corresponding gains for Ref. [11] are computed using Eq. (36). Figure 7a shows the variation of initial course angle χ_{d_0} and gain $\hat{k}_s$ with k_s . Percentage reduction in maximum curvature gained by using the proposed method as against Ref. [11] is plotted in Fig. 7b. Figure 7c corresponds to the percentage reduction in control effort (which is defined as integral of squared acceleration over given time interval $\int_0^t a^2 dt$, where lateral acceleration $a = V_g \dot{\chi}$) by using the proposed method. Figures 7b and 7c highlight that by using the proposed method the maximum curvature and the total control effort can be reduced significantly ($>70\%$ for $k_s > 4 \times 10^{-3}$). In terms of settling time (time taken to reach within $\pm 20 \text{ cm}$ of the desired path), this improvement in performance comes at a cost. Figure 7d plots the percentage increase in settling time for the proposed method. It can be seen that the increase in settling time is less than 30% for $k_s > 4 \times 10^{-3}$. In particular, for high values of gain k_s , significant reduction in maximum curvature and control effort can be achieved with a small increase in settling time.

B. Case 2: Circular Path Following

A circular path of desired radius $r_d = 50 \text{ m}$ is considered. As in case 1, the ground speed of the UAV is considered as $V_g = 10 \text{ m/s}$ and the guidance gain $k_c = 0.005$. The controller gains are chosen as $k_\chi = 50$ and $k_{\dot{\chi}} = 7.07$. Consider a UAV location outside the circular path at an initial distance of $r_0 = 100 \text{ m}$ with an initial orientation of $\gamma_0 = 225^\circ$. Figure 8 illustrates the simulation results. Figure 8a shows the UAV trajectory leading to the desired circular path with the corresponding radial error reducing to zero, as shown in Fig. 8b. The course angle profile, wrapped to vary between $[-\pi, \pi]$, is shown in Fig. 8c, where the commanded χ_d is tracked with negligible error. Figure 8d shows the variation of curvature with radial distance.

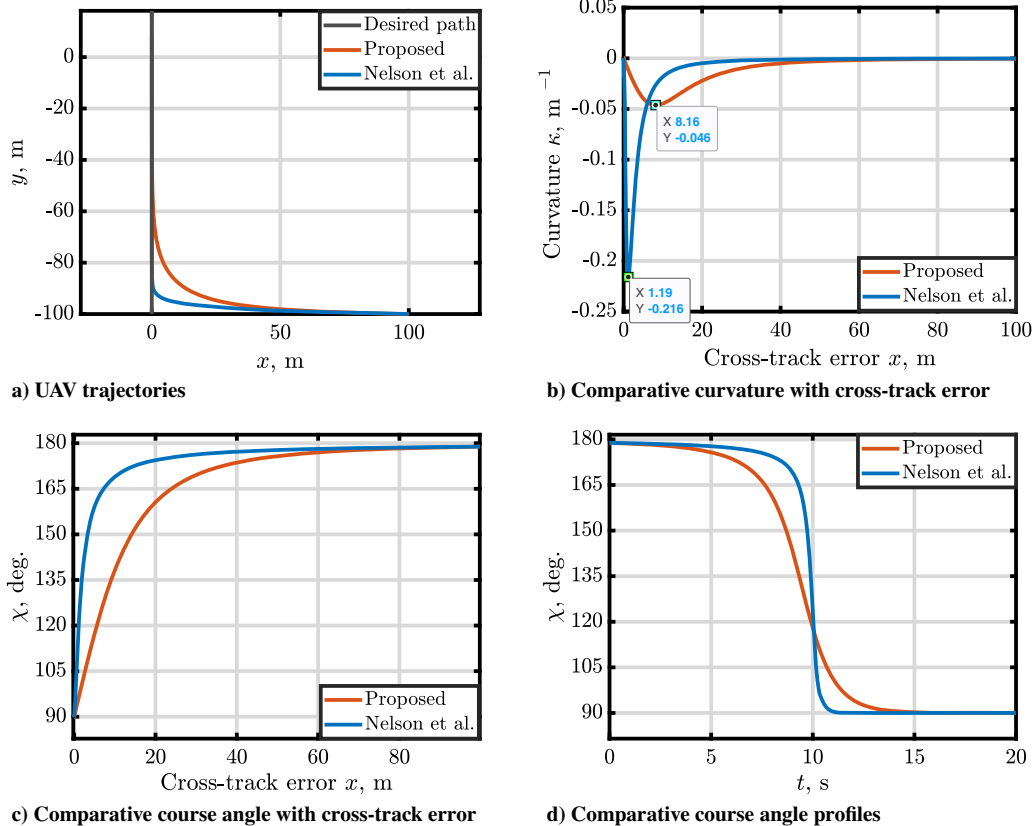


Fig. 6 Results for case 1: comparison with Ref. [11] for guidance gain $k_s = 0.005$.

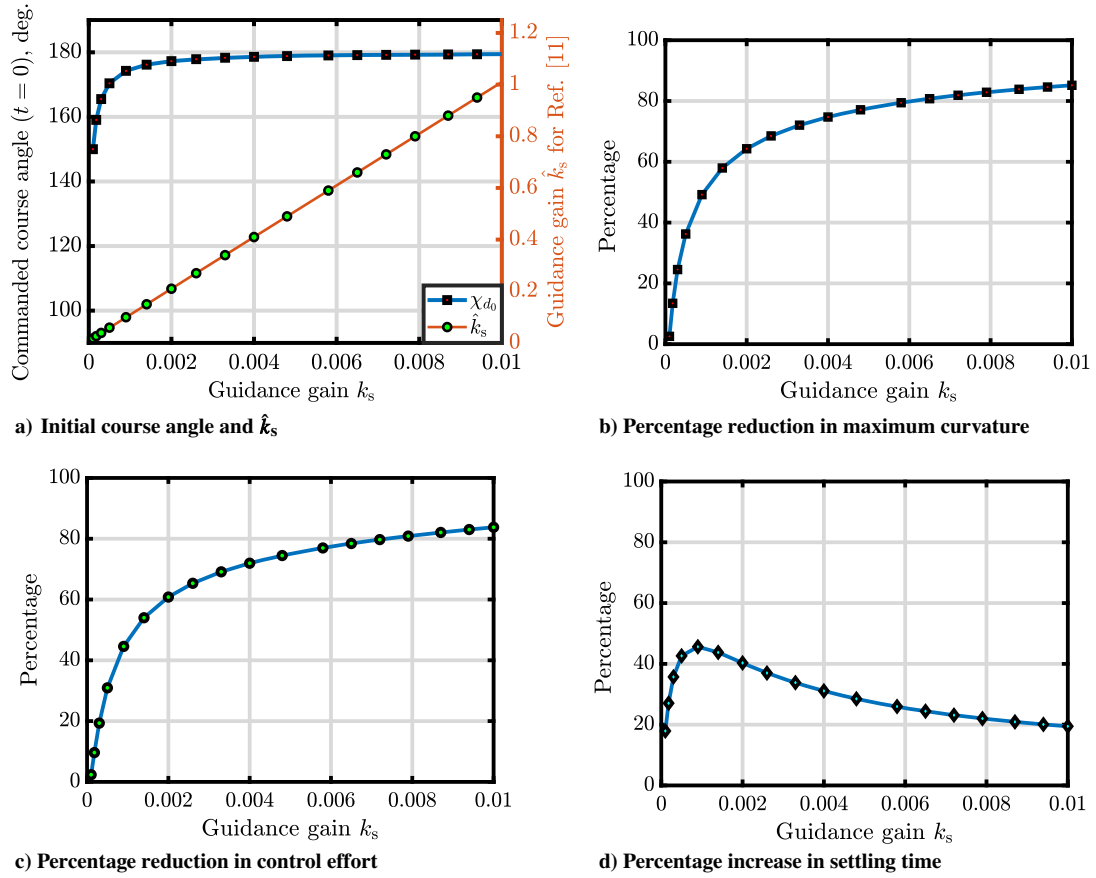


Fig. 7 Results for case 1: comparative study for a wide range of guidance gain k_s with Ref. [11].

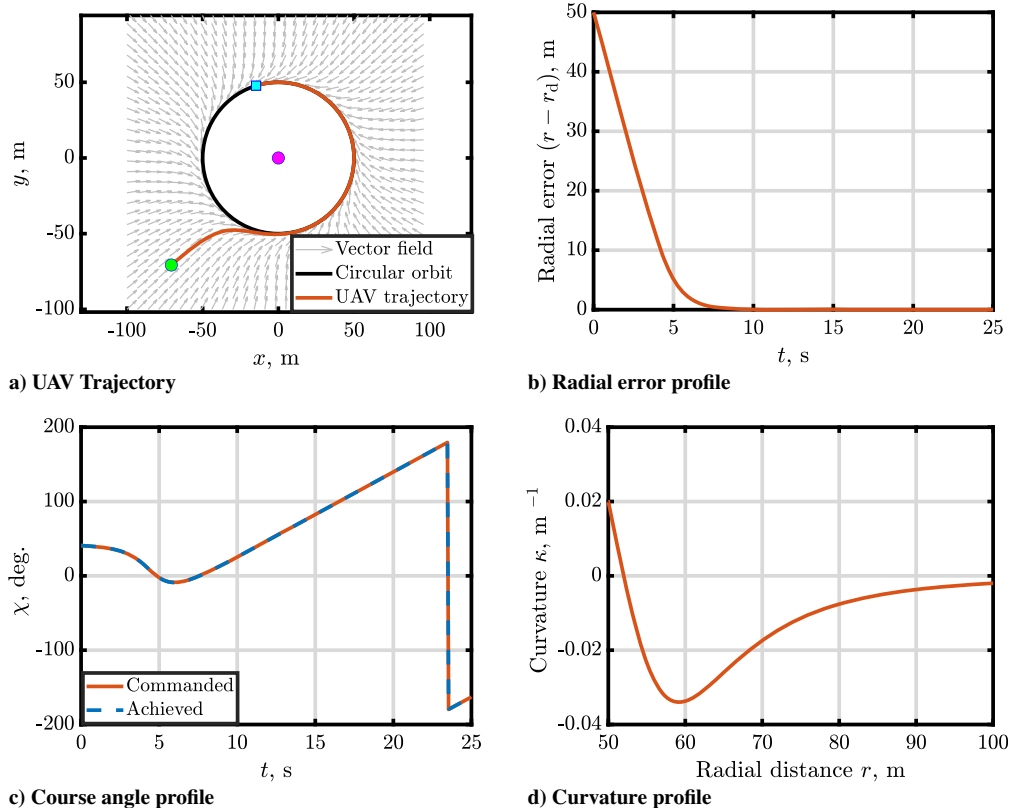


Fig. 8 Results for case 2: circular path following.

Additionally, the proposed circular path following method is compared with Ref. [11], for which the commanded course angle is given by

$$\hat{\chi}_d = \gamma + \frac{\pi}{2} + \tan^{-1}(\hat{k}_c(r - r_d)) \quad (39)$$

where $\hat{k}_c$ is the guidance gain. Considering $r_0 > r_d$, the gain $\hat{k}_c$ is obtained on equating the desired initial course angle given in Eqs. (23) and (39) as

$$\hat{k}_c = \sqrt{k_c^2(r_0 - r_d)^2 + 2k_c} \quad (40)$$

Consider that a UAV is initially located outside the desired circular orbit of radius $r_d = 50$ m at $(x_0, y_0) = (0 \text{ m}, -150 \text{ m})$ with $k_c = 0.005$. This choice of k_c , using Eq. (40), corresponds to $\hat{k}_c = 0.510$. The desired initial course angle obtained on substituting $k_c = 0.005$ in Eq. (23) for the proposed method and $\hat{k}_c = 0.510$ in Eq. (39) for

Ref. [11] is $\chi_{d_0} = \chi_0 = 88.88$ deg. The controller gains are chosen as $(k_\chi, k_\gamma) = (300, 17.32)$. Speed of the UAV is considered as $V_g = 12$ m/s. Figure 9 illustrates the comparative results. UAV trajectories are shown in Fig. 9a, and the corresponding curvature variation with radial distance is plotted in Fig. 9b. It is evident from Fig. 9b that the maximum curvature for the proposed method is $|\kappa_m| = 0.034 \text{ m}^{-1}$ at $r = 59.22$ m, which is considerably lower than $|\kappa_m| = 0.199 \text{ m}^{-1}$ at $r = 51.22$ m obtained for Ref. [11]. For a detailed comparison between the two vector field methods, simulations are carried out for a vast range of guidance gains in the interval $k_c \in [0.1, 10] \times 10^{-3}$. The corresponding guidance gains $\hat{k}_c$ for Ref. [11] are computed using Eq. (40). The variation of initial course angle χ_{d_0} and gain $\hat{k}_c$ with k_c is plotted in Fig. 9c. Figure 9d illustrates reduction (in percentage) of the maximum curvature as compared to that obtained using Ref. [11]. The corresponding percentage reduction in control effort is plotted in Fig. 9e. Figure 9f plots the percentage increase in settling time (time taken to reach within ± 20 cm. of the desired path) for the proposed method. For $k_c \in (0, 1.317] \times 10^{-3}$, the proposed

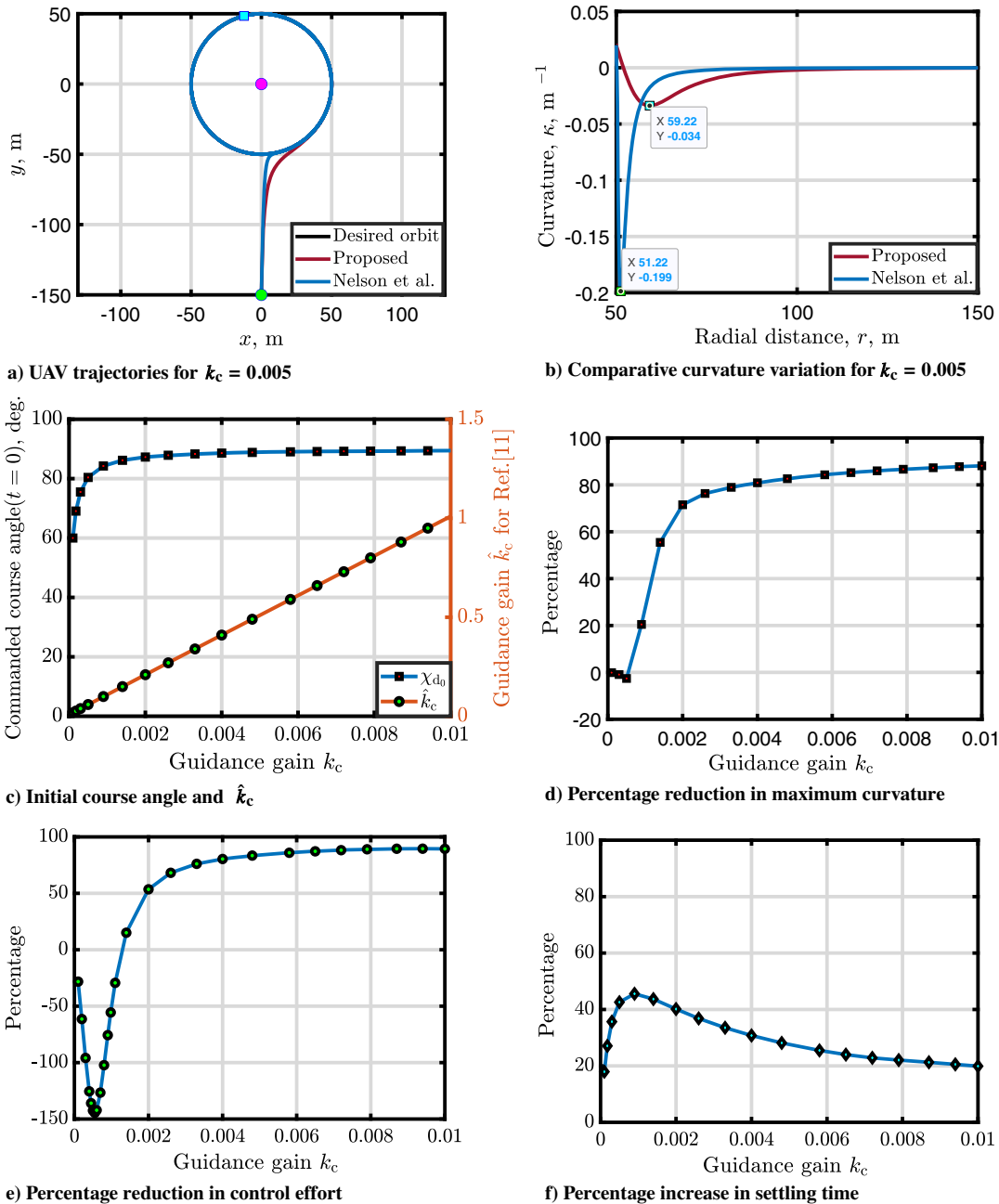


Fig. 9 Results for case 2: comparative study with Ref. [11].

method uses a higher control effort and a higher settling time. However, these trajectories correspond to long spiraling paths to the circle, which may not be of significant practical importance. It is noteworthy to highlight that the proposed method entails significant decrease in maximum curvature and control effort ($> 80\%$ for $k_c > 4 \times 10^{-3}$) at a cost of modest increase in settling time ($< 30\%$ for $k_c > 4 \times 10^{-3}$).

C. Case 3: Simulation Studies in the Presence of Wind

The presence of wind leads to deviation of ground speed V_g and course angle χ from airspeed V_a and heading angle ψ . Hence to accomplish tracking of the desired ground speed and the desired course angle, an airspeed autopilot is used in this case study.

With respect to wind triangle as shown in Fig. 10, the kinematic equations of motion that relate ground speed V_g , course angle χ , airspeed V_a , and heading angle ψ can be written as

$$V_g \cos \chi = V_a \cos \psi + w_x \quad (41)$$

$$V_g \sin \chi = V_a \sin \psi + w_y \quad (42)$$

where w_x and w_y are components of wind velocity V_w along x and y directions, respectively. For a prescribed ground speed V_{gd} and desired course angle χ_d , the corresponding desired airspeed can be obtained using Eqs. (41) and (42) as

$$V_{ad} = \sqrt{V_{gd}^2 + w_x^2 + w_y^2 - 2V_{gd}(w_x \cos \chi_d + w_y \sin \chi_d)} \quad (43)$$

To track V_{ad} , a first-order airspeed control [18] is considered as

$$\dot{V}_a = k_v(V_{ad} - V_a) \quad (44)$$

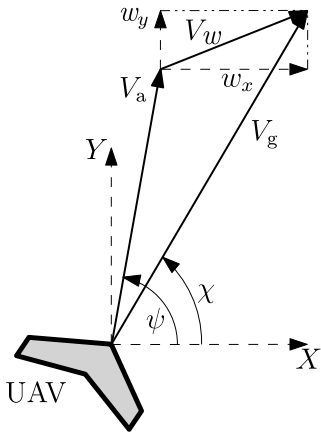


Fig. 10 Wind triangle.

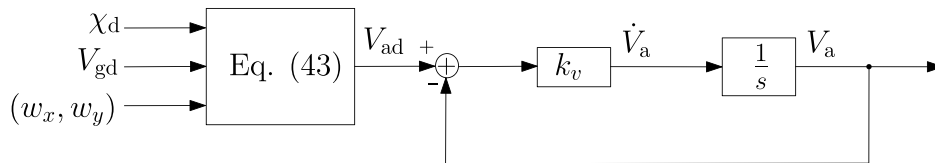


Fig. 11 Autopilot block diagram of airspeed-hold.

where k_v is a positive gain. The autopilot block diagram for airspeed control is shown in Fig. 11.

Since the autopilot considered in Eq. (33) directly controls χ , it can be used for validation in the presence of wind. Simulations are performed considering wind speed as $(w_x, w_y) = (-1, 2)$ m/s. The desired ground speed for straight-line path following is considered as $V_{gd} = 10$ m/s, and for circular path following it is considered as $V_{gd} = 12$ m/s. Other simulation parameters are listed in Table 2. Figure 12 illustrates the simulation results. Figure 12a shows the UAV trajectory following the desired straight-line path. UAV trajectories for the circular path are shown in Fig. 12b. The course angle profile for straight-line path following is shown in Fig. 12c. Figure 12d shows the course angle profiles for circular path following cases. UAV ground speed profiles for straight-line path and circular path scenarios are shown in Figs. 12e and 12f, respectively.

Further, we conducted 1000 Monte Carlo simulation runs using a uniform distribution of wind $(w_x, w_y) \sim U(-2, 2)$ m/s and parameters listed in Table 2. The tracking errors for the desired paths were very small, measuring less than 0.01 m for both the straight and circular paths, while the errors in tracking ground speeds were less than 0.01 and 0.013 m/s, respectively. The results show that the proposed guidance law is effective in guiding the UAV to follow desired straight and circular paths in the presence of wind.

D. Case 4: General Curvature Path

This case study extends the proposed path following logic for following curved paths given by $y = f(x)$, where y is an explicit function of x . We follow the approach given in Ref. [12] for extending a vector field developed for straight-line and circular path following to a general curvature path. Consider a UAV instantaneous position $(x_u(t), y_u(t))$ as shown in Fig. 13. A reference point $P(x_u(t), f(x_u(t)))$ is chosen on the curve and accordingly a tracking error is computed as $\epsilon = f(x_u(t)) - y_u(t)$. The course angle at P is identical to the slope at that point, $\chi_p = \tan^{-1}(dy/dx)$. To guide the UAV onto the path, the commanded course angle χ_d is formulated as

$$\chi_d = \begin{cases} \chi_p - \chi_o, & \epsilon \leq 0 \\ \chi_p + \chi_o, & \epsilon > 0 \end{cases} \quad (45)$$

where the proposed shaping function χ_o controls the smooth variation of UAV course angle as

$$\chi_o = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{1 + k_g \epsilon^2}\right) \quad (46)$$

Here, k_g denotes a positive guidance gain. It can be readily seen from Eq. (46) that χ_o lies in the range $[0, \pi/2]$ and varies smoothly for $\epsilon \in (-\infty, \infty)$. Also, at $\epsilon = 0$, $\chi_o = 0$ leads to the commanded course angle $\chi_d = \chi_p$, which is the desired direction along the path. For a very large lateral-tracking error ϵ , $\epsilon \gg 0$, $\chi_o \rightarrow \pi/2$, which leads to $\chi_d \rightarrow \chi_p + \pi/2$, a direction toward the path. Similarly, for $\epsilon \ll 0$,

Table 2 Parameters for path following in the presence for wind

Path	Initial position, (x_0, y_0) , m	Guidance gain	Initial airspeed (V_{a_0}) , m/s	Course-hold gains, (k_χ, k_χ)	Airspeed-hold gain (k_v)
Straight line	(100, -100)	$k_s = 0.002$	8	(30, 5.48)	5
Circular	(-7.07, -7.07) (inside) (-70.71, -70.71) (outside)	$k_c = 0.005$	10	(80, 8.94)	10

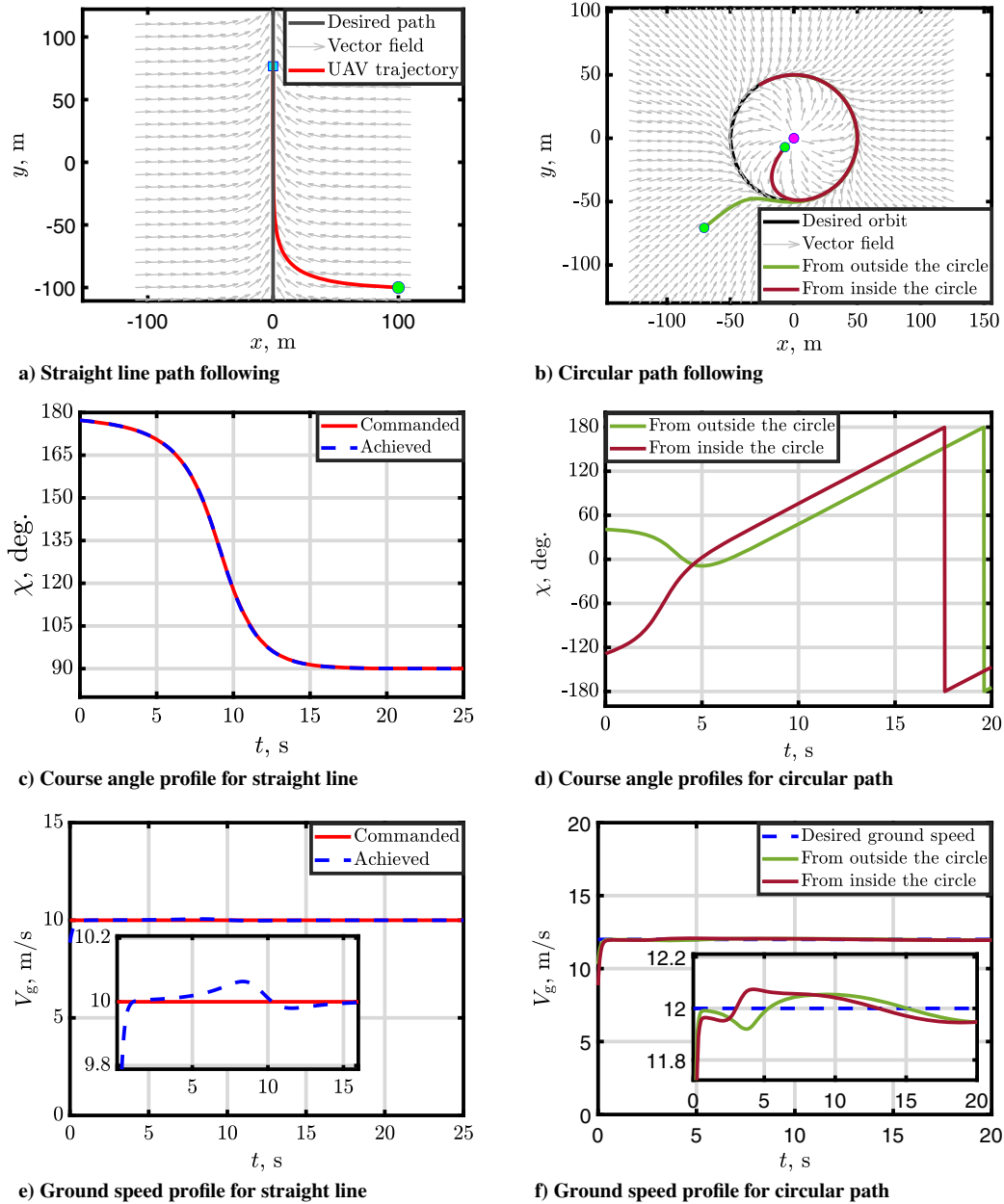


Fig. 12 Results for case 3: path following in the presence of wind.

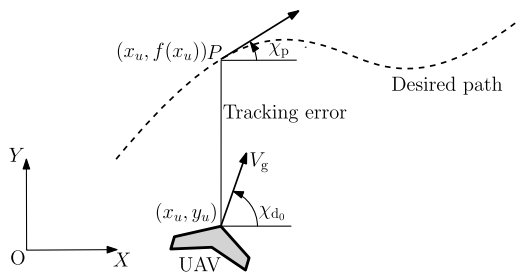


Fig. 13 General curvature path following geometry.

Table 3 Parameters for general curvature path following

Path	Initial position, (x_0, y_0) , m	Guidance gain, k_g	Initial course angle (χ_0) , deg	Course-hold gains, $(k_\chi, k_\ddot{\chi})$
Parabolic	$(-100, -80)$	0.002	17.56	$(50, 7.07)$
Sinusoidal	$(-90, -60)$	0.005	98.10	$(50, 7.07)$

$\chi_d \rightarrow \chi_p - \pi/2$. In this simulation study, we consider two different paths: sinusoidal and parabolic as governed by $y = 5 \sin(0.06x)$ and $y = 0.015x^2$, respectively. Table 3 lists parameters used in these simulations. Figures 14a and 14b show UAV trajectories leading to the respective paths. Corresponding curvature profiles are shown in Figs. 14c and 14d.

VI. Conclusions

Using an arcsine shaping function of the position error, this Note introduces a vector field guidance method for UAV path following. Two main scenarios are considered, namely, straight-line and circular paths, and respective guidance commands are proposed, which are easy to implement and computationally efficient. Simulations are carried out considering a second-order course-hold autopilot and a first-order airspeed control. The work includes simulation studies that compare the proposed method with a popular existing vector field, demonstrating that the proposed method reduces the maximum curvature and control effort required, with only a modest increase in the time taken to settle onto the desired path. The proposed method is also shown to be robust in the presence of wind, making it applicable

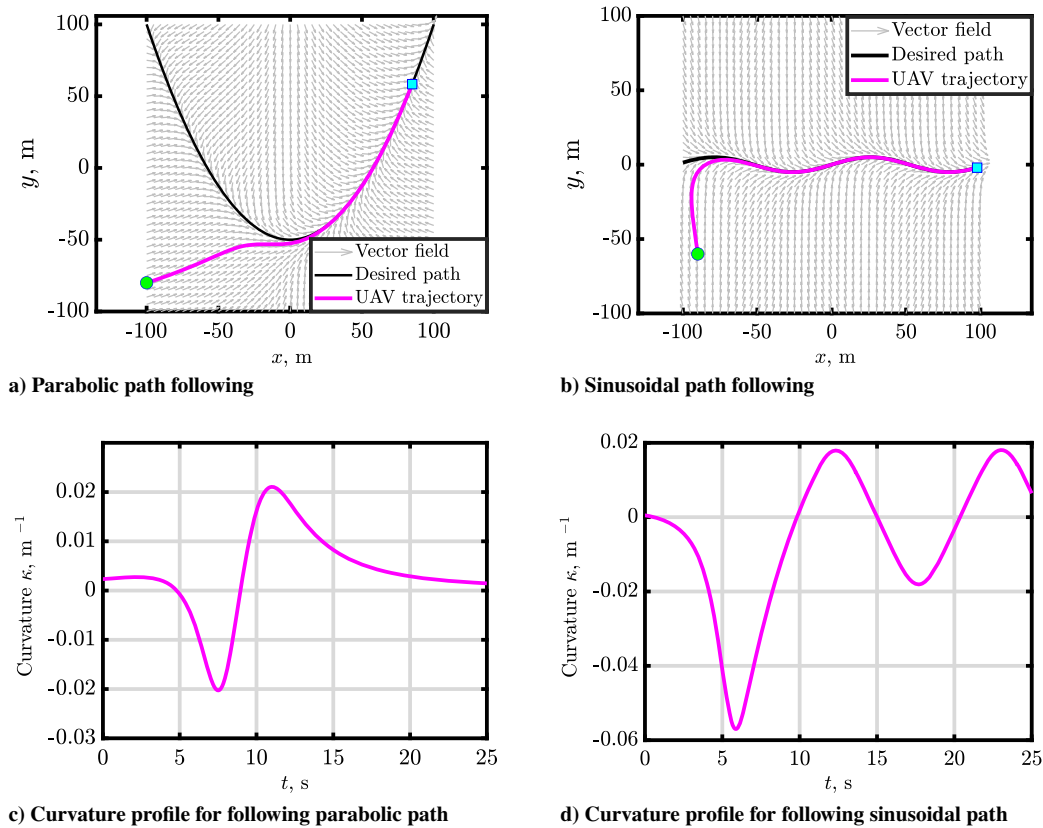


Fig. 14 Results for case 4: general curvature paths.

to real-world scenarios. Additionally, the work explores extending the proposed guidance methodology to follow paths with more complex curvature, such as parabolic and sinusoidal paths. Unlike the guidance logic that governs the UAV lateral acceleration, the proposed method directly commands the heading direction, and hence it is readily applicable to other platforms such as ground and marine vehicles.

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